

CFA OBSERVING TIME REQUEST

Proposal ID: 2835

Program No.: _____

P. I.: Christopher Stubbs

Semester 2026B (Jul 7 - Jan 16)

Email: cstubbs@cfa.harvard.eduCol.: A. Gil (Harvard), J. Medoff (Harvard), A. Villar (Harvard), R. SimcoeCol.: (MIT), Q. Wang (MIT), A. Clocchiatti (UC Chile), R. Cartier (UA Chile),Col.: A. Piro (Carnegie), Y. Dong (Harvard), J. Prieto (IEA), K. YadavalliCol.: (Harvard), A. Nugent (Harvard).Program Title: Testing Evolving Dark Energy with Rubin Supernovae.Telescope: Magellan BaadeInstrument: LLAMAS Integral Field Spect.Inst. Config.: Llamas has no configurable parameters.Nights Requested: D 0.5 G 2.0 B 0.5Queued Observing: Yes X No Type: new X continue engineering ToO Additional nights Student Project: research exam thesis X other Acceptable Dates: Aug-DecDates You Cannot Use: JulyTargets: Number 29 Magnitude Range 18.0 <= r <= 21.5Targets: RA Range 00:30:00 to 10:00:00 Dec Range -50:00:00 to 03:00:00

Abstract Include a short summary of your project. The summary should outline the main goals of the project and should not exceed 100 words.

We propose to use the Llamas integral field spectrograph to observe type Ia supernovae in the Rubin deep drilling fields, as well as high-precision white dwarf spectrophotometric standards. Our goal is to test the recent indications of time-variation in the dark energy equation of state parameter w . The ability of the Llamas instrument to begin integrations immediately after a slew, with no slit losses, makes it very well-suited this project. This proposal is part of a broader "Magnet" consortium where we are collaborating on the use of Magellan for followup observations of Rubin transients.

Testing Evolving of Dark Energy using Rubin Supernovae.

1 Scientific Justification

Identifying the mechanism behind the dark energy is arguably the most pressing open problem in fundamental physics. The only viable path forward at present is to exploit high-precision astronomical observations of the cosmic expansion history. Recent claims (Adame et al. 2025) from the DESI collaboration, where large scale structure measurements are combined with other constraints, indicate that rather than being constant the dark energy density could be evolving with time. If true, this would rule out Einstein’s cosmological constant as the source of dark energy, leaving us with an imperfect cancellation of quantum mechanical vacuum fluctuations, or some kind of phase transition in a cosmic field.

Type Ia supernovae remain an incisive probe of the history of cosmic expansion. By measuring their luminosity distance we know how long the light has been traveling, and by measuring their redshifts we can see how much the Universe has expanded in the intervening time. The Rubin observatory has now reliably started generating alerts, and the suite of “data brokers” has begun to parse and classify the resulting light curves. We can, with assurance, look forward to a steady stream of targets and to having well-calibrated Rubin photometry. The type Ia supernovae in the six Rubin deep-drilling fields will have high cadence multiband light curves from which we can extract a luminosity distance. Figure 1 shows the remarkable qualitative change in the number of type Ia’s that Rubin will produce, compared to today (Ivezic et al. 2019; The LSST DESC 2018).

Our goal is to test the hypothesis that dark energy evolves over time. Figure 2 shows the observable difference, as a function of redshift, between the standard Λ CDM cosmology and the evolving-dark-energy scenario suggested by DESI.

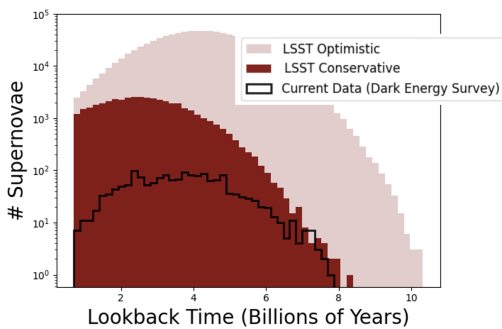


Figure 1: Number of Type Ia supernovae before and after Rubin. We will have our pick of targets, and we will focus on obtaining spectrophotometry of type Ia’s in elliptical hosts that are at peak.

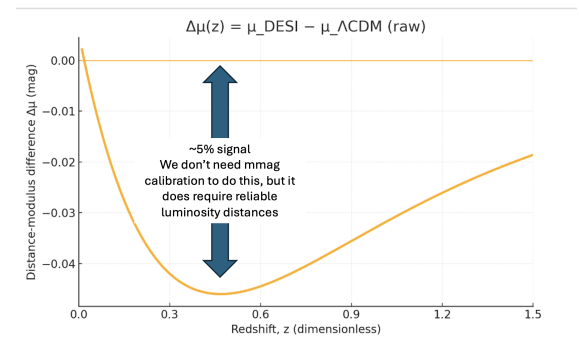


Figure 2: Difference between SN distance moduli for the DESI evolving dark energy scenario vs. more standard Λ CDM. There is a predicted multi-percent photometric signal that is strongest at modest redshifts.

There is no realistic prospect of obtaining spectroscopy for all the Rubin deep drilling field type Ia supernovae. We will rely on photometric redshift determinations of the host galaxies, and light curve phenomenology to discriminate Ia's from other cosmic transients. But those characterizations will need to be tested and errors quantified using spectral measurements of a subset of the objects. The ability of Llamas to obtain actual *spectrophotometry* of faint type Ia's is a new development, and one that we intend to fully exploit. Our group has laid Boyd et al. (2026) the groundwork for a network of faint white dwarf spectrophotometric standards that we will incorporate into our observing program. We aspire to calibrate the spectrophotometry of supernovae out to z of 0.4 to the few-mmag level, which is unprecedented. **We will produce the world's first catalog of high-precision spectrophotometric measurements of type Ia supernovae over the redshift range relevant to test the DESI hypothesis.**

Specific measurement objectives include:

- Testing the proposal that dark energy has evolved between $z=0.5$ and today,
- Validating the photo- z measurements of host galaxy redshifts,
- Testing the photometric determination of Ia redshifts from their light curves,
- Assessing the dependence of Hubble diagram residuals on host galaxy properties.

2 Experimental Design

We intend to obtain single-epoch spectrophotometry of type Ia supernovae and their host galaxies, using the Llamas integral field spectrograph on Baade. The redshift range of interest ($0.1 < z < 0.4$, see Figure 2) corresponds to peak Ia r band magnitudes of 18.5 to 22. The network of faint white dwarf spectrophotometric standards that our team has established in anticipation of Rubin SNe will play a central role in our observing campaign. These stars have SEDs whose relative flux vs. wavelength is known at the few-mmag level, and this will allow us to precisely compensate for both atmospheric and instrumental throughput.

The rate of Rubin alerts, along with the high-cadence coverage of the deep drilling fields and prompt difference photometry, will provide us with our choice of objects near peak, on any given night. This will allow us to make maximum use of allocated Magellan time, in contrast with prior Ia cosmology programs where sparse target lists drove us to often take spectra well away from peak brightness.

We are asking for a combination of bright, dark, and gray time that we will use to obtain a statistically meaningful sample of supernova and host galaxy spectra. The bright time will concentrate on the brighter, nearby objects. Figure 3 shows the result from the Llamas exposure time calculator for a 1200 sec integration on a 20th magnitude type Ia supernova (spectrum is shown unredshifted). We expect to easily reach SNR of 10 to 40 per pixel in a 20 min exposure, with enough spectral resolution to work between the OH emission lines. We will need to spectrally bin for our highest redshift objects, at $r=22$ nd magnitude, in a reasonable exposure. We expect we can attain $\text{SNR} > 10$ for $\lambda < 800\text{nm}$ at 22nd mag in binned

spectra with 2 hours of integration time. The intrinsic width of the Doppler-broadened SN spectral features means we can do this without compromising effective spectral resolution.

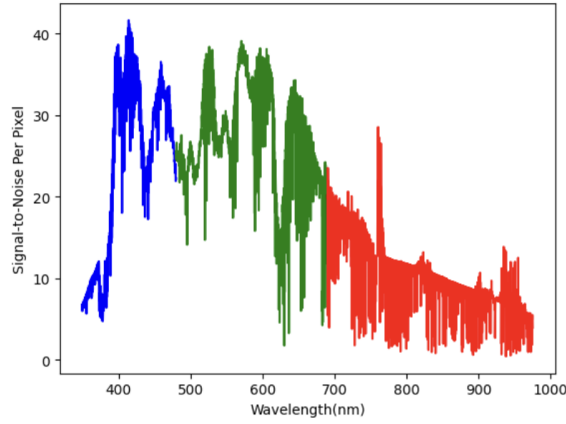


Figure 3: Result from Llamas exposure time calculator for a 20th magnitude type Ia SN, 1200 sec exposure. This includes dark time sky noise.

For this initial phase of our program, our tentative breakdown of observations is shown in Table 1. This assumes one minute of slew/overhead per observation. Most of the time we’ll be moving within a 3.5 degree Rubin deep drilling field, which will be highly efficient.

<i>Redshift Bin (z)</i>	<i>Moon Phase</i>	<i>Peak r</i>	<i>Binning</i>	<i>Exp. (min)</i>	<i>$N_{targets}$</i>
<i>0.10 – 0.20</i>	<i>Bright</i>	<i>19.3</i>	<i>2×</i>	<i>35</i>	6
<i>0.20 – 0.30</i>	<i>Grey</i>	<i>20.6</i>	<i>2×</i>	<i>95</i>	6
<i>0.30 – 0.35</i>	<i>Grey</i>	<i>21.3</i>	<i>4×</i>	<i>160</i>	4
<i>0.30 – 0.35</i>	<i>Dark</i>	<i>21.3</i>	<i>4×</i>	<i>45</i>	2
<i>0.35 – 0.40</i>	<i>Dark</i>	<i>21.9</i>	<i>4×</i>	<i>180</i>	1
<i>Standards</i>	<i>All</i>	<i>18.0</i>	<i>1×</i>	<i>1</i>	<i>10</i>
<i>Total Targets</i>					29

Table 1: Balanced 3-night observing plan for Magellan/Baade. The schedule provides 6 SNe Ia per bin for $z < 0.35$. Overheads are 1 minute per target. The total time allocation is 30 hours (5h Dark, 20h Grey, 5h Bright). We use a combination of lunar phase-dependent sky noise and spectral binning to achieve our goals. Exposure times are per object.

By focusing on type Ia’s in dust-free elliptical hosts, with high cadence multiband Rubin photometry we can expect a per-SN distance modulus uncertainty of $\sigma_{mu} \sim 0.08$ mag. With 6 objects per redshift bin we can attain a bin-averaged distance modulus uncertainty per redshift bin of 0.04 mag. If we compare this to the 0.4 mag predicted discrepancy in distance modulus shown in Figure 2, we should achieve a high statistical confidence in testing the DESI evolving dark energy hypothesis.

Observations of the white dwarf precision spectrophotometric calibration sources will be interleaved, and obtained across an appropriate range of redshifts.

3 Addendum

Progress Report:

We expect for Rubin transient followup observations to become a long-term program at Magellan, for which this is our initial proposal. We have not used Llamas before, but the instrument PI Rob Simcoe is a collaborator who will support the observations and data reduction.

Description of Collaboration:

MAGNETS, a collaboration between several Magellan partner institutions, is a brand new effort motivated by the start of LSST alerts in March of this year. Therefore, the proposed work constitutes the first ever use of the Magellan telescopes to follow up LSST transients. This proposal will contribute allocated nights to the MAGNETS pool of time with the remaining partners proposing to their own TACs to do transient science. We note that science cases across the MAGNETS collaboration vary significantly and the collaboration exists because of a shared need for a queue based observational strategy which is not currently otherwise available at Magellan. The science case outlined in this proposal is distinct from other proposals that will be submitted at other Magellan partner institutions.

Our collaboration's plan is to pool our awarded observing time and develop an internal queue schedule to address a wide range of science goals. Among the participants Stubbs' group has taken the lead on using type Ia supernovae for observational cosmology. Other collaboration member's primary interests are in core collapse objects, or exotic transients. We will apportion the queue observing time and target prioritization in proportion to the time awarded by the respective TACs, with the full data set being accessible to all. Student projects will be declared and protected appropriately.

Student Thesis/Project Description:

Graduate students Akum Gil and Jonah Medoff are Harvard students in the physics PhD program, with Stubbs as thesis advisor. They will be working on all aspects of this project, including creating a systematic uncertainty budget for increased precision in luminosity distance determination, and associating Hubble diagram residuals vs. host galaxy attributes. They will also contribute to the collaborative infrastructure needed to execute this blended program and to distribute the data.

Previous Time Awards:

Although I have been a Magellan user in the past, including runs on IMACS, LDSS, and PISCO, the last time was more than 5 years ago. Those observations were used to obtain redshifts, weak lensing mass estimates, and velocity dispersions of galaxy clusters detected by the South Pole Telescope, and we were very successful in generating numerous science papers from the data collected.

Key Publications:

These papers illustrate our effective exploitation of prior Magellan time allocations, including a major slit mask spectroscopy campaign and weak lensing imaging observations

of SPT clusters. High and Ruel were graduate students in Stubbs’ group, and Stalder and Bayliss were postdocs.

- Ruel, J., Bazin, G., Bayliss, M., et al. 2014, *Optical Spectroscopy and Velocity Dispersions of Galaxy Clusters from the SPT-SZ Survey*, ApJ, 792, 45
- High, M. D., Stalder, B., Stark, A. A., et al. 2012, *Weak-Lensing Mass Measurements of Five Galaxy Clusters in the South Pole Telescope Survey*, ApJ, 758, 68
- Bocquet, S., Dietrich, J. P., Schrabback, T., et al. 2019, *Cluster Cosmology Constraints from the 2500 deg² SPT-SZ Survey: Inclusion of Weak Lensing Data from Magellan and HST*, ApJ, 878, 55
- Bocquet, S., Saro, A., Mohr, J. J., et al. 2015, *Mass Calibration and Cosmological Analysis of the SPT-SZ Galaxy Cluster Sample Using Velocity Dispersion and Y_x Data*, ApJ, 799, 214
- Capasso, R., Saro, A., Mohr, J. J., et al. 2019, *Galaxy Cluster Mass Calibration from a Joint Analysis of STAGES, SPT, and Magellan Velocity Dispersion Data*, MNRAS, 482, 1043

Target List:

We will be focusing on type Ia supernovae detected in the Rubin deep drilling fields. Those are listed in Table 1, along with an appropriate subset of our high-precision spectrophotometric standard white dwarf stars.

Table 2: Rubin Deep Drilling Field (DDF) Centers and Selected Spectrophotometric Calibrators

<i>Name</i>	<i>RA</i>	<i>Dec</i>	<i>r</i>	<i>source</i>	<i>Visibility</i>
<i>ELAIS-S1 (Field)</i>	<i>00:37:48</i>	<i>-44:01:12</i>		<i>DDF</i>	<i>Aug-Oct</i>
<i>WD 2311+174</i>	<i>23:14:24</i>	<i>+17:44:45</i>	<i>18.2</i>	<i>Standard</i>	<i>Aug-Nov</i>
<i>XMM-LSS (Field)</i>	<i>02:22:50</i>	<i>-04:45:00</i>		<i>DDF</i>	<i>Sep-Nov</i>
<i>WD 0308-565</i>	<i>03:10:31</i>	<i>-56:21:03</i>	<i>17.5</i>	<i>Standard</i>	<i>Oct-Dec</i>
<i>ECDFS (Field)</i>	<i>03:32:31</i>	<i>-28:06:00</i>		<i>DDF</i>	<i>Oct-Dec</i>
<i>EDFS-a (Field)</i>	<i>03:55:36</i>	<i>-49:19:12</i>		<i>DDF</i>	<i>Oct-Dec</i>
<i>WD 0548-001</i>	<i>05:51:15</i>	<i>-00:07:10</i>	<i>18.0</i>	<i>Standard</i>	<i>Nov-Jan</i>
<i>COSMOS (Field)</i>	<i>10:00:26</i>	<i>+02:10:48</i>		<i>DDF</i>	<i>Jan-Mar*</i>
<i>WD 1034+001</i>	<i>10:37:01</i>	<i>+00:13:38</i>	<i>18.2</i>	<i>Standard</i>	<i>Jan-Apr*</i>

**Note: COSMOS is poorly placed for the July–December window and is included here as a reference for a complete DDF list.*

Bibliography:

- Adame, A. G., et al. 2025, *DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations*, JCAP, 2025, 001
- Ivezic, Z., Kahn, S. M., Tyson, J. A., et al. 2019, *LSST: From Science Drivers to Reference Design and Anticipated Data Products*, ApJ, 873, 111

- The LSST Dark Energy Science Collaboration, Abell, R., et al. 2018, *LSST Dark Energy Science Collaboration (DESC) Science Requirements Document*, arXiv, 1809, 01669
- Boyd, B. M., Saha, A., Olszewski, E. W., et al. 2026, *Attaining Spectral Energy Distributions With Sub-Percent Uncertainties: All-Sky DA White Dwarf Spectrophotometric Standard Stars For Large Telescopes And Surveys*, AJ, 171, 265

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